Aatman Pandya

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Summary

AI engineer building production conversational agents: long-term memory, retrieval with reranking, schema-enforced LLM extraction and safety, on FastAPI, Postgres/pgvector and the OpenAI, Anthropic and Voyage APIs. Architected and built the long-term memory of a health-tech AI agent now used by real beta users.

Education

BITS Pilani, Hyderabad Campus — Hyderabad, India
M.Sc. (Hons) Biological Sciences & B.E. (Hons) Mechanical Engineering — *Expected May 2027*

Experience

AI Engineering Intern (Practice School II) — Jul 2026 – Dec 2026
DXFactor — ELO, a conversational AI agent for relationship intelligence (health tech) — *Ahmedabad*

- **Architected and built ELO's long-term memory from the ground up**, now live with beta users: background extraction of facts, people and open conflicts from every chat; layered recall over past chats; and a working window with a running summary. Built ELO's first version end to end (FastAPI, Next.js, Postgres + pgvector, Redis, arq).
- Proposed and built a **Voyage reranker** over each user's nearest past chats to replace a fixed embedding-distance cut-off, raising how often the right past chat reaches the prompt from **72% to 92%** of messages; added back-reference and time cues and multi-turn carry-over.
- Built the **eval suites that gate every memory and RAG change**: test users with up to 240 facts and 120 chats run through the production prompt path, graded by an **LLM judge**, with prompt-cache cost tracking.
- Built **RAG over clinician-approved content packs** (chunked retrieval, ranking, an ingest gate for approved versions); technique reach rose from **0% to 93%** with 0/10 false fires and no change across 718 regression scenarios.
- Made memory extraction **schema-enforced** (OpenAI, Anthropic) with attribution and durability guards so the model can't store invented facts, and built a safety classifier that handles **romanised-Hindi and code-mixed** input.
- Shipped DPDP-compliant erasure with per-user encryption keys, rate limits, and log redaction that keeps user data out of database errors.

AI Engineering Intern — Summer 2026
Zenalyst AI — voice-AI calling platform — *Remote*

- Added **speech-to-speech** calling (OpenAI Realtime, Gemini Live) to a LiveKit Agents platform that previously ran only an STT → LLM → TTS cascade.
- Moved the call LLM to Groq with Anthropic fallback, added per-call token accounting, a fuzzy-match response cache, and **tool-calling** agents that send confirmation emails after booking, reschedule and payment calls.

Data Analytics Intern, Octalsoft — Python reporting pipelines over 1,000+ customer records; Oracle APEX modules — May 2024 – Jul 2024

Projects

Multilingual Film Search | *Python, BM25, dense embeddings, Wikidata, Wikipedia* — 2026

- Built search over 9,274 Indian films for English, native-script, Hinglish and misspelt queries, from a catalogue assembled with Wikidata SPARQL and batched Wikipedia plot extraction.
- Wrote a 300-query graded test set split into dev and test before any search code existed; field-weighted BM25 with transliteration and phonetic keys raised dev **nDCG@10 from 0.47 to 0.78**. Dense and hybrid retrieval are being benchmarked.

Subscription Churn Model in Production | *XGBoost, DuckDB, MLflow, FastAPI, Docker* — 2026

- Predicted churn for 993K KKBox subscribers from 21.5M transactions and 392M listening-log rows with leak-free cut-offs; XGBoost **ROC-AUC 0.899**, 0.873 on a later out-of-time cohort; top decile holds 67.6% of churners.
- Shipped it with an MLflow registry, a FastAPI scoring service in Docker, and drift monitoring that caught a real calibration gap (6.5% predicted vs 9.0% actual), fixed by recalibration (ECE 0.025 → 0.001).

Technical Skills

AI / LLM: LLM agents, long-term memory, RAG, reranking, embeddings, structured extraction, tool calling, speech-to-speech, LLM safety, LLM evals (LLM-as-judge, regression suites, nDCG)
Models & APIs: OpenAI (GPT, Realtime), Anthropic Claude, Google Gemini, Groq, Voyage embeddings and rerank, LiveKit Agents
ML: XGBoost, scikit-learn, calibration, drift monitoring, MLflow, Pandas, NumPy, DuckDB
Engineering: Python, TypeScript, SQL, Java · FastAPI, Next.js, PostgreSQL + pgvector, Redis, arq, Celery, Alembic, Docker, Pytest, Git